Proof that the Levi-Civita Tensor is covariantly constant under the Levi-Civita connection

Goal

We show that the Levi-Civita tensor $\varepsilon_{\mu\nu\rho\sigma}$ is covariantly constant under the Levi-Civita connection, i.e.,

$$\nabla_\lambda \varepsilon_{\mu\nu\rho\sigma} = 0.$$

Some Key Points

- It is important to understand the difference between the *Levi-Civita symbol* $\tilde{\varepsilon}_{\mu\nu\rho\sigma}$ and the *Levi-Civita tensor* $\varepsilon_{\mu\nu\rho\sigma}$.
- The Levi-Civita *symbol* $\tilde{\varepsilon}$ picks up a Jacobian transformation factor under a coordinate transformation:
$$\tilde{\varepsilon}_{\mu'_1\cdots\mu'_n} = \det\left(\frac{\partial x^\alpha}{\partial x^{\mu'}}\right)\frac{\partial x^{\beta_1}}{\partial x^{\mu'_1}}\cdots\frac{\partial x^{\beta_n}}{\partial x^{\mu'_n}}\tilde{\varepsilon}_{\beta_1\cdots\beta_n}$$
- Clearly, this doesn't follow the tensor transformation law. The actual tensor that can be constructed in place of this is the **Levi-Civita tensor**, denoted $\varepsilon_{\mu\nu\rho\sigma}$, which is related to the Levi-Civita symbol as:
$$\epsilon_{\mu\nu\rho\sigma} = \sqrt{-g}\,\tilde{\epsilon}_{\mu\nu\rho\sigma}$$
- The Levi-Civita *tensor* $\varepsilon_{\mu\nu\rho\sigma}$ defines the concept of an **oriented volume** element. In 4D, the volume element is hence defined as:
$$\omega = \varepsilon_{\mu\nu\rho\sigma}\, dx^\mu \otimes dx^\nu \otimes dx^\rho \otimes dx^\sigma$$
- If ε is covariantly constant, it means parallel transport preserves the infinitesimal oriented volume locally under the Levi-Civita connection.

Derivation

We note:

$$\nabla_\lambda \varepsilon_{\mu\nu\rho\sigma} = \nabla_\lambda\big(\sqrt{-g}\,\tilde{\varepsilon}_{\mu\nu\rho\sigma}\big) = \big(\nabla_\lambda\sqrt{-g}\big)\,\tilde{\varepsilon}_{\mu\nu\rho\sigma}$$

Step 1: Find $\nabla_\lambda \sqrt{-g}$

Consider first $\partial_\lambda g_{\mu\nu}$.

Using Jacobi's formula for the determinant:

$$\partial_\lambda g = g\, g^{\mu\nu}\, \partial_\lambda g_{\mu\nu} \tag{i}$$

Contracting with $g^{\mu\nu}$:

From the Levi-Civita connection (metric compatibility), we have:

$$\partial_\lambda g_{\mu\nu} - \Gamma^\alpha_{\lambda\mu}\, g_{\alpha\nu} - \Gamma^\alpha_{\lambda\nu}\, g_{\mu\alpha} = 0$$

Contracting both sides with $g^{\mu\nu}$:

$$g^{\mu\nu}\, \partial_\lambda g_{\mu\nu} - \Gamma^\alpha_{\lambda\mu}\, \delta^\mu_\alpha - \Gamma^\alpha_{\lambda\nu}\, \delta^\nu_\alpha = 0$$

$$g^{\mu\nu}\, \partial_\lambda g_{\mu\nu} = 2\, \Gamma^\alpha_{\lambda\alpha} \tag{ii}$$

(The last two terms are non-zero only for $\mu = \nu = \alpha$, and hence they're equal.)

Substituting (ii) into (i):

$$\partial_\lambda g = g \cdot 2\, \Gamma^\alpha_{\lambda\alpha} \tag{iii}$$

Using (iii) and (ii) together gives:

$$\partial_\lambda \sqrt{-g} = \frac{1}{2\sqrt{-g}} \cdot (-\partial_\lambda g) = \frac{1}{2\sqrt{-g}} \cdot (-g) \cdot 2\, \Gamma^\alpha_{\lambda\alpha} = \sqrt{-g}\; \Gamma^\alpha_{\lambda\alpha}$$

$$\boxed{\partial_\lambda \sqrt{-g} = \sqrt{-g}\; \Gamma^\alpha_{\lambda\alpha}} \tag{iv}$$

Step 2: Compute $\partial_\lambda \varepsilon_{\mu\nu\rho\sigma}$

Since $\varepsilon_{\mu\nu\rho\sigma} = \sqrt{-g}\; \tilde{\varepsilon}_{\mu\nu\rho\sigma}$ and the Levi-Civita *symbol* $\tilde{\varepsilon}$ is a constant (its components are just ± 1 or 0 in any coordinate system):

$$\partial_\lambda \varepsilon_{\mu\nu\rho\sigma} = \left(\partial_\lambda \sqrt{-g}\right) \tilde{\varepsilon}_{\mu\nu\rho\sigma}$$

Using (iv):

$$\partial_\lambda \varepsilon_{\mu\nu\rho\sigma} = \Gamma^\alpha_{\lambda\alpha} \left(\sqrt{-g}\; \tilde{\varepsilon}_{\mu\nu\rho\sigma}\right) = \Gamma^\alpha_{\lambda\alpha}\; \varepsilon_{\mu\nu\rho\sigma} \tag{v}$$

Step 3: Compute the covariant derivative $\nabla_\lambda \varepsilon_{\mu\nu\rho\sigma}$

Since $\varepsilon_{\mu\nu\rho\sigma}$ is a rank-4 covariant tensor, its covariant derivative picks up one connection term per index:

$$\nabla_\lambda \varepsilon_{\mu\nu\rho\sigma} = \partial_\lambda \varepsilon_{\mu\nu\rho\sigma} - \Gamma^\alpha_{\lambda\mu}\,\varepsilon_{\alpha\nu\rho\sigma} - \Gamma^\alpha_{\lambda\nu}\,\varepsilon_{\mu\alpha\rho\sigma} - \Gamma^\alpha_{\lambda\rho}\,\varepsilon_{\mu\nu\alpha\sigma} - \Gamma^\alpha_{\lambda\sigma}\,\varepsilon_{\mu\nu\rho\alpha} \tag{vi}$$

Substituting (v) into (vi):

$$\nabla_\lambda \varepsilon_{\mu\nu\rho\sigma} = \Gamma^\alpha_{\lambda\alpha}\,\varepsilon_{\mu\nu\rho\sigma} - \left(\Gamma^\alpha_{\lambda\mu}\,\varepsilon_{\alpha\nu\rho\sigma} + \Gamma^\alpha_{\lambda\nu}\,\varepsilon_{\mu\alpha\rho\sigma} + \Gamma^\alpha_{\lambda\rho}\,\varepsilon_{\mu\nu\alpha\sigma} + \Gamma^\alpha_{\lambda\sigma}\,\varepsilon_{\mu\nu\rho\alpha}\right)$$

By Einstein summation, the sum of the four connection terms on the right is precisely $\Gamma^\alpha_{\lambda\alpha}\,\varepsilon_{\mu\nu\rho\sigma}$ (since ε is totally antisymmetric and summing over α in each slot saturates the trace):

$$\Rightarrow \quad \nabla_\lambda \varepsilon_{\mu\nu\rho\sigma} = \Gamma^\alpha_{\lambda\alpha}\,\varepsilon_{\mu\nu\rho\sigma} - \Gamma^\alpha_{\lambda\alpha}\,\varepsilon_{\mu\nu\rho\sigma}$$

$$\therefore \quad \boxed{\nabla_\lambda \varepsilon_{\mu\nu\rho\sigma} = 0} \qquad \blacksquare$$